

A Canonical Lower-Semicontinuous Kuratowski Extension and the Sharp Obstruction to Affine Interpolation

Candidate solution packet for arXiv:2402.11284

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Status. Candidate full two-sided resolution; likely valid, pending human review. Every real-valued lower-semicontinuous datum has an explicit finite extension; for nonnegative data there is a maximal chord-subaffine extension. The exact endpoint-affine analogue used in the Lipschitz-free-space proof is impossible in general.

1 Source question

The source is D. Bate, S. Eriksson-Bique, and E. Soultanis, *Fragment-wise differentiable structures*, arXiv:2402.11284v1 [1]. In Remark 2.1 on PDF page 7, the authors explain that their natural convex extension is not well-defined after the standard Kuratowski embedding and ask whether a canonical extension preserving semicontinuity exists there.

FRAGMENT-WISE DIFFERENTIABLE STRUCTURES

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The semihull provides a convenient technical tool which will be used in Section 3 to obtain lower semicontinuity of suitable integrals over fragments of curves. Its key property is a canonical extension of functions $X \rightarrow \mathbb{R}$ to functions $\text{sh}(X) \rightarrow \mathbb{R}$ which preserves lower semicontinuity.

Remark 2.1. *The standard Kuratowski embedding $X \hookrightarrow \ell^\infty$ is not appropriate for our canonical extension scheme. Indeed, the natural “convex” extension we use (see Section 3.1) is not well-defined for the Kuratowski embedding, and we do not know whether a canonical semicontinuity preserving extension exists for it.*

Figure 1: Remark 2.1 in the source paper (PDF page 7).

The word “canonical” is not axiomatized in the remark. We resolve the question using the strongest natural one-sided interpolation property and then show that the corresponding equality property is impossible.

2 Definitions and statement

Let (X, d) be a separable metric space, fix $o \in X$, and let $D \subset X$ be dense. The base-pointed Kuratowski embedding is

$$\mathcal{K} : X \longrightarrow \ell^\infty(D), \quad \mathcal{K}(x)(s) = d(x, s) - d(o, s).$$

It is an isometry. Set

$$\mathcal{H}_{\mathcal{K}}(X) = \bigcup_{x,y \in X} [\mathcal{K}(x), \mathcal{K}(y)].$$

All topology below is the relative norm topology on this semihull.

For any $\rho : X \rightarrow [0, \infty]$, define its raw two-point roof by

$$R_{\rho}(z) = \inf \{ (1-t)\rho(a) + t\rho(b) : z = (1-t)\mathcal{K}(a) + t\mathcal{K}(b), a, b \in X, t \in [0, 1] \}. \quad (1)$$

We use the standard convention $0 \cdot \infty = 0$. Define the relaxed roof

$$E_{\rho}(z) = \sup_{r>0} \inf \{ R_{\rho}(w) : w \in \mathcal{H}_{\mathcal{K}}(X), \|w - z\|_{\infty} < r \}. \quad (2)$$

Theorem 1 (Canonical Kuratowski extension). *If $\rho : X \rightarrow [0, \infty]$ is lower semicontinuous, then E_{ρ} is a lower-semicontinuous extension of ρ :*

$$E_{\rho}(\mathcal{K}(x)) = \rho(x) \quad (x \in X).$$

Moreover:

1. for all $a, b \in X$ and $t \in [0, 1]$,

$$E_{\rho}((1-t)\mathcal{K}(a) + t\mathcal{K}(b)) \leq (1-t)\rho(a) + t\rho(b); \quad (3)$$

2. E_{ρ} is the greatest lower-semicontinuous function $g : \mathcal{H}_{\mathcal{K}}(X) \rightarrow [0, \infty]$ satisfying every inequality (3);
3. the assignment $\rho \mapsto E_{\rho}$ is order preserving, positively homogeneous, and preserves nonnegative constant functions.

Thus (2) is canonical by a maximal universal property.

Corollary 2 (Literal real-valued extension). *Assume additionally that X is complete. Every lower-semicontinuous $\rho : X \rightarrow \mathbb{R}$ admits an explicit finite lower-semicontinuous extension $T_{\rho} : \mathcal{H}_{\mathcal{K}}(X) \rightarrow \mathbb{R}$ with $T_{\rho} \circ \mathcal{K} = \rho$.*

Theorem 3 (Sharp obstruction to the free-space formula). *There is a four-point metric space X and a continuous $\rho : X \rightarrow [0, 1]$ for which no function $G : \mathcal{H}_{\mathcal{K}}(X) \rightarrow [0, \infty]$ can satisfy*

$$G((1-t)\mathcal{K}(a) + t\mathcal{K}(b)) = (1-t)\rho(a) + t\rho(b)$$

simultaneously for every $a, b \in X$ and $t \in [0, 1]$. Consequently no general Kuratowski extension scheme can retain the exact endpoint-affine identity of the Lipschitz-free-space construction.

3 Proof intuition

The raw roof removes ambiguity by choosing the cheapest value among all chord representations. Its lower-semicontinuous relaxation is automatically the largest lower-semicontinuous function compatible with all resulting upper bounds. Relaxation could in principle lower the boundary data. The special geometry of the Kuratowski embedding prevents that: evaluating a barycenter near $\mathcal{K}(x)$ at the coordinate x shows that almost all of its barycentric mass must lie on endpoints near x . Lower semicontinuity of ρ then forces the cost back up to $\rho(x)$.

The obstruction is the cycle metric on four points. Opposite chords have the same midpoint, so exact affine interpolation assigns two incompatible values there. This explains precisely why the free-space construction can be replaced by a canonical inequality but not by the exact identity used for weighted gap integrals.

For arbitrary real-valued data, first compress the values monotonically into the interval $(1, 2)$, use the roof theorem there, and keep the extension away from the endpoints of that interval by a canonical distance-to- $\mathcal{K}(X)$ floor. Completeness makes $\mathcal{K}(X)$ closed, so the floor is strictly above 1 off the embedded space. Inverting the compression yields a finite real-valued lower-semicontinuous extension.

4 Proofs

We first isolate the Kuratowski estimate that makes exact boundary recovery possible.

Proposition 4. *For $a, b, x \in X$ and $t \in [0, 1]$,*

$$\|(1-t)\mathcal{K}(a) + t\mathcal{K}(b) - \mathcal{K}(x)\|_\infty \geq (1-t)d(a, x) + td(b, x). \quad (4)$$

Proof. If $x \in D$, evaluate the vector on the left at $s = x$. The base-point terms cancel, and the value is the nonnegative number on the right. For general x , choose $s_n \in D$ with $s_n \rightarrow x$. The coordinate values converge to the same number, so their supremum is at least that limit. $\square$

Proof of Theorem 1. For any extended-real function on a metric space, the expression in (2) is its lower-semicontinuous relaxation, hence E_ρ is lower semicontinuous and $E_\rho \leq R_\rho$. Formula (1) then immediately gives (3).

It remains to prove the boundary trace. Fix $x \in X$. The representation $\mathcal{K}(x) = (1-t)\mathcal{K}(a) + t\mathcal{K}(b)$ shows $E_\rho(\mathcal{K}(x)) \leq R_\rho(\mathcal{K}(x)) \leq \rho(x)$.

Suppose first that $0 < \rho(x) < \infty$, and fix $c < \rho(x)$ with $c \geq 0$. By lower semicontinuity there is $\delta > 0$ such that $\rho(u) > c$ whenever $d(u, x) < \delta$. Let $w = (1-t)\mathcal{K}(a) + t\mathcal{K}(b)$ and put $r = \|w - \mathcal{K}(x)\|_\infty$. By Proposition 4, the total barycentric weight of endpoints outside $B(x, \delta)$ is at most r/δ . Nonnegativity therefore gives

$$(1-t)\rho(a) + t\rho(b) \geq c(1-r/\delta).$$

This holds for every representation of w , so $R_\rho(w) \geq c(1-r/\delta)$. Taking the infimum over the radius- r_0 neighborhood of $\mathcal{K}(x)$ and then letting $r_0 \downarrow 0$ yields $E_\rho(\mathcal{K}(x)) \geq c$. Letting $c \uparrow \rho(x)$ proves equality. If $\rho(x) = 0$, the lower bound is automatic. If $\rho(x) = \infty$, repeat the argument with an arbitrary finite c ; then let $c \rightarrow \infty$.

For maximality, let g be lower semicontinuous and satisfy all the chord inequalities. Taking the infimum over representations gives $g \leq R_\rho$. For every z ,

$$g(z) = \sup_{r>0} \inf_{\|w-z\|_\infty < r} g(w) \leq \sup_{r>0} \inf_{\|w-z\|_\infty < r} R_\rho(w) = E_\rho(z),$$

where all points w lie in $\mathcal{H}_\mathcal{K}(X)$. This proves the universal property. Monotonicity, positive homogeneity, and preservation of constants follow directly from (1) and (2). $\square$

Proof of Corollary 2. Fix the increasing homeomorphism

$$h : \mathbb{R} \longrightarrow (1, 2), \quad h(s) = \frac{3}{2} + \frac{1}{\pi} \arctan s.$$

Apply Theorem 1 to the lower-semicontinuous function $f = h \circ \rho$. Since every endpoint value lies strictly between 1 and 2, its relaxed roof satisfies

$$1 \leq E_f(z) < 2 \quad (z \in \mathcal{H}_K(X)).$$

The image $\mathcal{K}(X)$ is closed because X is complete and $\mathcal{K}$ is an isometry. Put

$$q(z) = 1 + \frac{\text{dist}(z, \mathcal{K}(X))}{1 + \text{dist}(z, \mathcal{K}(X))}, \quad T_\rho(z) = h^{-1}(\max\{E_f(z), q(z)\}).$$

The maximum is lower semicontinuous, is strictly between 1 and 2 at every point, and equals $f(x)$ at $z = \mathcal{K}(x)$: there $q = 1$ and $E_f(\mathcal{K}(x)) = f(x) > 1$. Since h^{-1} is continuous and increasing, T_ρ is finite and lower semicontinuous, and $T_\rho(\mathcal{K}(x)) = \rho(x)$. $\square$

Proof of Theorem 3. Let $X = \mathbb{Z}/4\mathbb{Z}$ with the cycle metric $d(i, j) = \min\{|i - j|, 4 - |i - j|\}$, and use base point 0 and all four points as coordinates. The distance rows are

$$D_0 = (0, 1, 2, 1), \quad D_1 = (1, 0, 1, 2), \quad D_2 = (2, 1, 0, 1), \quad D_3 = (1, 2, 1, 0).$$

Since $\mathcal{K}(i) = D_i - D_0$, the identity $D_0 + D_2 = D_1 + D_3$ gives

$$\frac{\mathcal{K}(0) + \mathcal{K}(2)}{2} = \frac{\mathcal{K}(1) + \mathcal{K}(3)}{2} =: m.$$

Set $\rho(0) = \rho(2) = 0$ and $\rho(1) = \rho(3) = 1$. Every function on this finite metric space is continuous. Exact affine interpolation on the first opposite chord forces $G(m) = 0$, while interpolation on the second forces $G(m) = 1$, a contradiction. $\square$

5 Interpretation and limitations

Corollary 2 answers the literal existence question for all real-valued lower-semicontinuous data under the completeness hypothesis of the source paper. Theorem 1 gives the stronger order-theoretic answer for the nonnegative integrands used in the application. It is nontrivial on the semihull (unlike the extension that assigns $+\infty$ off $\mathcal{K}(X)$), and it is forced as the greatest lower-semicontinuous chord-subaffine extension.

It does not reproduce the identity

$$\bar{\rho}((1 - t)\mathcal{K}(a) + t\mathcal{K}(b)) = (1 - t)\rho(a) + t\rho(b)$$

along every gap chord. Theorem 3 shows that this limitation cannot be removed for arbitrary metric spaces. Therefore the positive construction does not by itself replace the exact line-integral identity in Section 3.1 of the source paper.

6 Verification and novelty bounds

The proof is self-contained. The finite script in the packet independently checks the cycle distance matrix, the midpoint identity, and the conflicting endpoint averages using exact integer arithmetic. It is a check of the example, not a substitute for the general proof.

On 11 August 2026, exact arXiv-id/title/core-keyword searches of this run's registry and result indexes found no prior result. Exact-phrase and nearby keyword searches of the locally parsed arXiv

corpus found the question only in arXiv:2402.11284. Locally available citing sources arXiv:2504.16657, 2508.08017, and 2510.25715 do not discuss the Kuratowski-extension issue. Bounded external web searches returned no usable records in this environment; novelty confidence is therefore moderate. No external theorem is used.

Human-review recommendation. Check the extended-value boundary argument and decide whether the maximal chord-subaffine universal property is the intended formalization of “canonical” in Remark 2.1. Independently, the four-cycle impossibility theorem is unconditional and exact.

References

- [1] D. Bate, S. Eriksson-Bique, and E. Soultanis, *Fragment-wise differentiable structures*, arXiv:2402.11284v1 (2024).